

Occupational and Environmental Cardiovascular Hazards and Diseases

Department of Occupational and
Environmental Medicine

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Pre-Class Assessment: Do You Know What "Overwork" Means?

- **Case Study 1: Mr. Wang**
- Security guard at a textile factory
- Works 10 hours/day, weekends off, 15 years of service
- Chronic hepatitis B, no smoking or drinking
- Recently assigned night shifts
- Liver function abnormal (GOT: 500, GPT: 480); mild fatty liver
Concerned that overwork may have caused liver damage

- In the past two months, he has often worked night shifts.

During a recent routine employee health checkup, his liver function tests showed abnormal results (GOT: 500, GPT: 480), and an ultrasound revealed mild fatty liver.

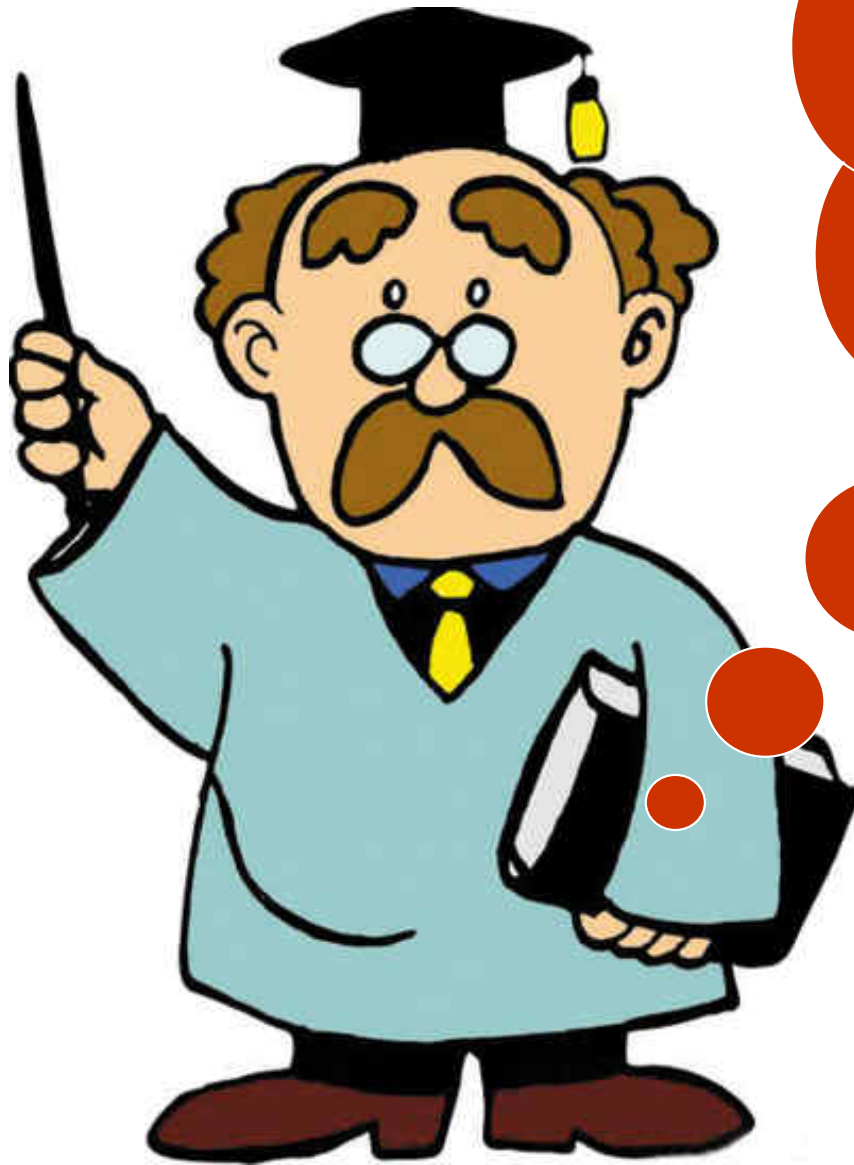
- Mr. Wang is worried that his condition might be caused by overwork — that he is literally “burning out his liver” due to work-related fatigue.

- **Case 2**

Title: Case 2: Ms. Lee – Stress and Stomach Problems

- **Content:**

- Secretary at a trading company, 8 years of service
- Regular 9-to-5 schedule, weekends off
- Recently involved in office relocation, often works until 7 PM
- Skips lunch due to workload; diagnosed with peptic ulcer
- Wonders if stress and irregular meals caused her condition



Let's think about
it...Do you think
the health
problems in these
cases were
caused by
overwork?

Course Outline

- Understanding Overwork: Historical evolution from Japan to Taiwan
- Principles for Recognizing Overwork-related Illness
- The Role of Responsibility Systems
- Case Studies and Practical Analysis
- Cardiovascular Toxic Chemicals in the Workplace

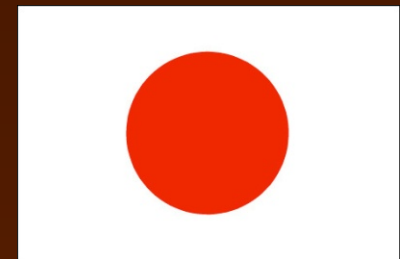
Learning from Japan: Historical Milestones

Decade	Milestone
1960s	Standards for CNS and circulatory diseases (1961)
1980s	Medical definition of Karoshi (death from overwork)
1990s	"Karoshi 110" campaign; first legal recognition of suicide due to overwork
2000s	Shift from "excessive workload" to "triggering illness" model
2010s	Expanded recognition criteria for occupational diseases

過勞死

- **What Is Karoshi?**

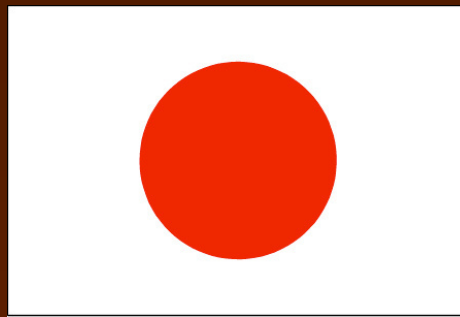
- Coined by Dr. Tetsunojo Uehata in the 1980s
- Refers to sudden death due to overwork-induced hypertension, arteriosclerosis, stroke, heart failure, arrhythmia
- Includes mental health conditions like depression and suicide



- **Legal Framework in Japan**

- **Occupational Safety and Health Act:** Defines occupational injuries
- **Labor Standards Act:** Specifies diseases covered by compensation
- **2010 Amendment:** Includes stroke, heart attack, subarachnoid hemorrhage, and aortic dissection caused by long working hours

- Overwork not only refers to a state of death, but also includes serious illnesses (such as "stroke"), as well as mental illnesses caused by overwork, such as overwork depression and even overwork suicide.



Legalization of Overwork in Japan: Criteria for Determining Occupational Accidents for Cardiovascular and Brain Diseases

Four revisions in 1995, 1996, and 2001 gradually relaxed the definition of overwork, including "excessive load."



Japanese Labor Time Regulations

- **Breaks:**

- 6 hours: 45 min break
- 8 hours: 1 hour break) 中場休息

- **Legal Holidays:**

- At least 1 day off per week or 4 days off every 4 weeks



After the death of an engineer...

TW
案例

- Led to revision of diagnostic guidelines
- New title: “Guidelines for Recognizing Occupationally Triggered Cerebrovascular and Cardiovascular Diseases (excluding trauma)”



Principles for Recognizing Overwork-related Illness

- Diseases are often personal but worsened by work conditions
- If work contributes $>50\%$ to disease progression, it may be classified as occupational
- Recognition includes listed diseases and those supported by autopsy or literature

表列疾病



Target Diseases 目標疾病

腦血管疾病

- **Cerebrovascular:** Stroke, cerebral infarction, subarachnoid hemorrhage, hypertensive encephalopathy

心血管疾病

- **Cardiac:** Myocardial infarction, acute heart failure, aortic dissection, arrhythmia, cardiac arrest, sudden cardiac death

Overtime Thresholds 加班時數

- Month before onset: ≥ 100 hours ^{92 (old)}
- 2–6 months before onset: ≥ 80 hours/month ^{擇一}
average _{92 (old)}
- Employers must provide evidence of workload

正常工時: 40 hr/week

- **Overtime hours:**

Normal working hours are 40 hours per week

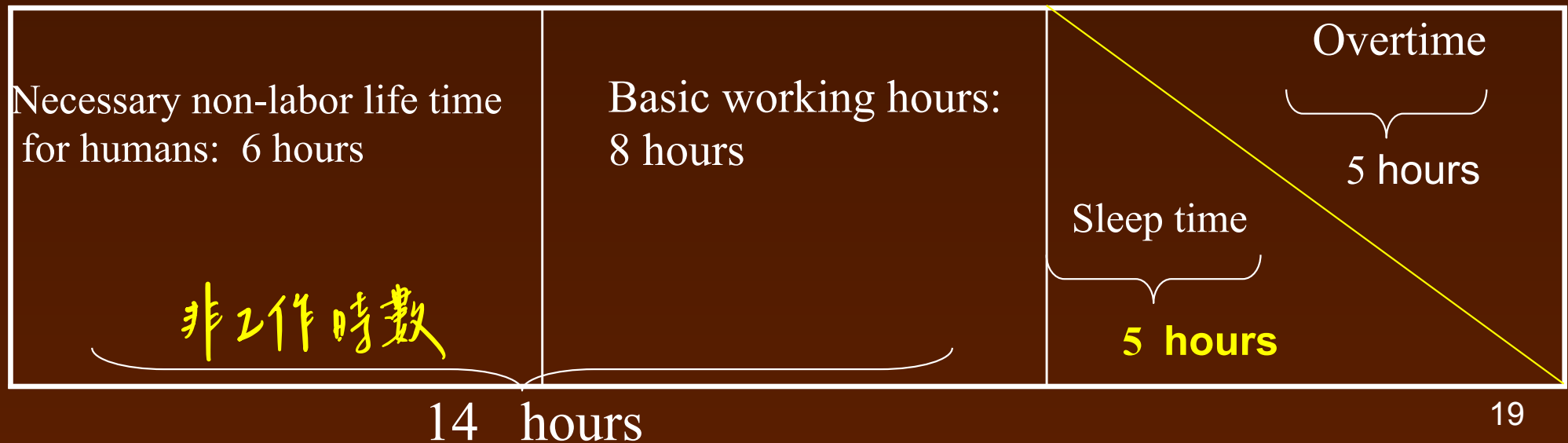
Any hours exceeding normal working hours are defined as **overtime hours**. ⇒ 超過 40 → 加班

- **Employer's burden of evidence principle:**
To avoid difficulties in determining workload differences between labor and management, first listen to the labor side's opinions and then ask the management side to provide evidence.

Sleep and Risk

- <5 hours sleep increases risk of brain/heart disease by 1.8–3.2× compared to 6–8 hours

(日本厚生労働省労働基準局安全衛生部 労働衛生課 土井 智史)



Workload Assessment Criteria

- Categories:
 - Abnormal events
 - Short-term excessive workload
 - Long-term excessive workload
- Factors:
 - Irregular hours, frequent travel, night shifts
 - Extreme temperatures, noise
 - Mental stress from high-risk or isolated tasks

The onset of overwork disease is not limited to the workplace.

Mental Stress Indicators

- High-risk responsibilities (e.g., life-threatening tasks)
- Time-sensitive or high-pressure assignments
- Conflict resolution, lack of support, complex development roles
- Exposure to traumatic events or major accidents

(一) 日常伴隨精神緊張的工作

具體的工作	評估觀點	
經常負責會威脅自己或他人生命、財產的危險性工作	危險性程度、工作量（勞動時間、勞動密度）、勤務期間、經驗、適應能力、公司的支援、預估的受害程度等	
有迴避危險責任的工作		
關乎人命、或可能左右他人一生重大判決的工作		
處理高危險物質的工作		
可能造成社會龐大損失責任的工作		
有過多或過分嚴苛的限時工作	勞動內容、困難度、強制性、有無懲罰	工作量（勞動時間、勞動密度）、勤務期間、經驗、適應能力、公司的支援等
需在一定的期間內（如交期等）完成的困難工作	阻礙因素的嚴重性、達成的困難度、有無懲罰、變更交期的可能性等	
負責處理客戶重大衝突或複雜的勞資紛爭	顧客的定位、損害程度、勞資紛爭解決的困難度等	
無法獲得周遭理解或孤立無援狀況下的困難工作	工作的困難度、公司內的立場等	
負責複雜困難的開發業務、或公司重建等工作	企劃案中所持立場、執行困難度等	

(二) 接近發病時期所伴隨的精神緊張之工作相關事件

事件	評估觀點
因職業災害造成嚴重受傷或疾病	職災受害的程度、有無後遺症、回歸社會的困難度等
與發生重大事故及災害直接相關	事故的大小、加害程度等
經歷（目擊）悲慘的事故或災害	事故及受害程度、恐懼感、異常性程度等
被追究重大事故（事件）的責任	事故（事件）的內容、責任歸咎情形、對社會負面影響的程度、有無懲罰等
工作上嚴重失誤	失敗的程度、重大性、損害等的程度、有無懲罰等
未達成限時工作內容	限時工作量的內容、達成的困難度、強制性、達成率的程度、有無懲罰等
工作異動（調職、人員調配、職務轉換、派任等）	工作內容、身份等的變動、異動理由、不利的程度等
與上司、客戶等產生重大紛爭	紛爭發生時的狀況、程度等

責任制

Responsibility System v.s. Overwork

- Legal responsibility system $\neq$ exemption from overwork regulations ↳ 責任制仍有過勞
- Must be approved by labor committees and government agencies
- Overtime pay still required

Approved Industries for Responsibility System

Industry	Example Roles
Finance	Bank managers
IT	System engineers
Legal	Legal advisors
Advertising	Creative directors
Aviation	Flight crew
Security	Guards, surveillance staff
Healthcare	Hospital staff
Childcare	Caregivers, social workers

- The legal process is completed only after approval by the Council of Labor Affairs' Labor Standards Advisory Committee, a determination of reasonable working hours by the Labor-Management Conference, and final approval by the county or city labor authorities.
- *This does not mean unlimited overtime work without overtime pay.*

合法責任制產業別	備註
事業單位之首長、主管以及獲有配車人員之駕駛。	
【金融業】 銀行業僱用之經理職以上人員	勞動基準法施行細則第五十條之一第一款規定者
【資訊服務業】 資訊服務業僱用之負責事業經營管理工作之主管人員，及系統研發工程師與維護工程師	
【法律服務業】 法律服務業僱用之負責事業經營管理工作之主管人員，以及法務人	
【廣告業】 廣告業僱用之經理級以上人員，以及創作人員	
【會計服務業】 會計服務業僱用之會計助理人員具會計師法第十二條規定之資格	
【航空服務業】 航空公司空勤組員(前艙與後艙工作人員)	
【保全業】 保全業之保全人員、電腦管制中心監控人員、經理級以上人員者	
【保險業】 保險業之外勤人身保險業務員依保險業務員管理規則領有登錄證者。	
【房屋仲介業】 房屋仲介業之不動產經紀人員(含業務主管人員)	
【醫療保健服務業】 含國軍醫院及其附設民眾診療處)之部分場所及人員	
【幼托業】 托兒所保育員以及社會福利服務機構之輔導員(含保育員、助理保	

After-class practical sharing

Mr. X, 35 years old, was an engineering quality control team leader at a technology factory in southern Taiwan.

...He was found dead suddenly at home on January 1, 2000.

He had been working in his current job since October 2009, primarily designing and repairing engineering pipelines. He had previously told his family that he suffered from high work pressure, irritability, and a poor appetite.

- From July 2009 to October 2009, he was on **unpaid leave** due to the economic downturn. ...Before July 2009, he worked as an engineer at a technology factory in central Taiwan.
- Current work hours (as reported by family members, not time clock records):

Leaves home at 7:00 AM and returns home at 7:00 PM
(round trip, approximately an hour each way) ⇒ 不算過勞時數

From 9:00 PM to 2:00 AM (next day), he organizes the day's work on his computer and emails it to the manager of the company for the next morning's meeting (a task assigned by the company manager).

- Lifestyle: No smoking or drinking.
- Past medical history: no history of diabetes, hypertension, hyperlipidemia or heart disease.

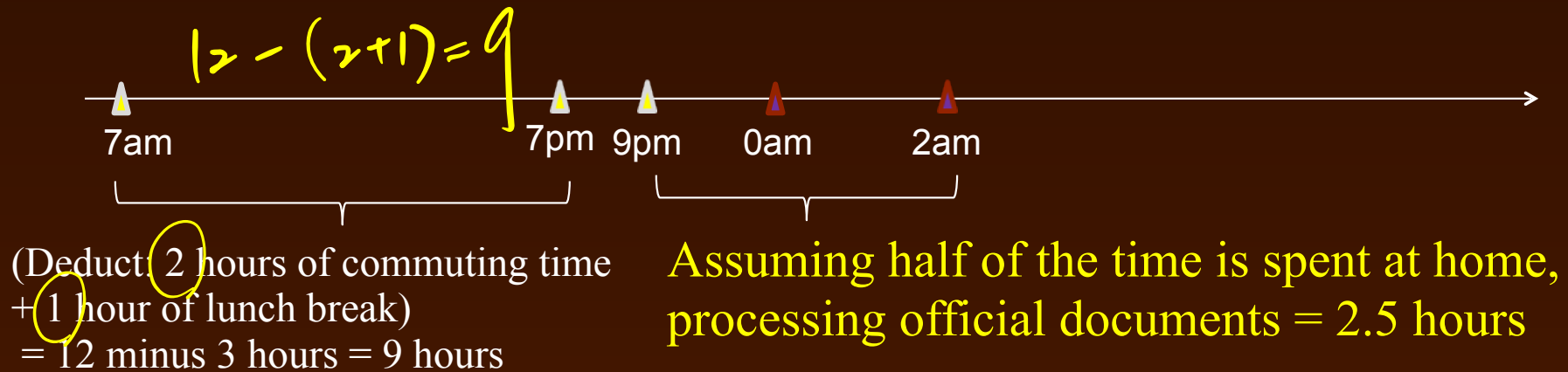
- [Case Study 1]

- How do you estimate actual work hours?

- Can work occur at home? A: 可以

- Is commuting time included in the calculation of work hours?

...Assume that the lunch break is 1 hour



➡ Estimated daily work-related hours = $9 + 2.5 = 11.5$ hours

➡ Monthly work-related hours = $11.5 \text{ hours/day} \times 22 \text{ days/month} = 253 \text{ hours/month}$
Monthly overtime hours = $253 \text{ hours} - \text{statutory monthly normal working hours} = 253 \text{ hours/month} - [(84 \text{ hours}/14 \text{ days}) \times 30 \text{ days/month}] = 253 \text{ hours/month} - 180 \text{ hours/month} = 73 \text{ hours/month}$

↪ 舊版每 week 工時 (2 weeks)



3.5 month
2009 unemployment period

2010

July	Aug.	Sep.	Oct.		Nov.	Dec.	Jan.	
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During the **working period** (October to January), the average monthly overtime hours were 73 hours:

Overtime hours in the month before the onset of illness (death) = 73 hours.

Average monthly overtime hours **2 to 6 months** before the onset of illness (death) (i.e., July to November 2009) = $(0 + 0 + 0 + 73 + 73) / 5 = 29.2$ hours/month

↳ < 72 hr, 無過勞 ⇒ 理論要通過 (7 month 平均)
(old version)



Is this a case of
burnout
(overwork) ?



Thank you for listening

